Executive Summary

Mechanistic Interpretability of Open Source LLMs. Reproduction of On the Biology of a Large Language Model.

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The standard test for whether a language model represents a concept is a linear probe, a classifier trained on the hidden states, whose high accuracy is read as evidence that the concept is encoded along a direction. This project asks whether the direction a probe reads is also the direction the model uses to compute its answer, and shows on a controlled task that it is not. The single notion of a concept direction separates into three geometrically and causally distinct objects, a reading direction along which the concept is decoded, a writing direction along which it is deposited into the residual stream, and a use direction through which it reaches the output.

The project began as a small-scale reproduction of the circuit tracing methodology of Ameisen et al. and the analytical programme of On the Biology of a Large Language Model on an open model, Qwen3-4B with pre-trained per-layer transcoders. Attribution graphs over transcoder features were constructed on four corpora and validated by targeted interventions, confirming that the methodology transfers, with compact multi-step chains whose nodes carry readable object-level content. The reproduction also exposed its own limits. Causal effects were carried by ensembles of features rather than by single necessary paths, the attribution labels of the graph did not predict the direction of a feature's causal effect, and many causally relevant features activated comparably on both classes. These observations motivated the second and main phase, the study of the residual stream directly at the level of directions.

The central task is a binary classification of nuclear decay type, in which the model answers alpha or beta from a class-balanced corpus of 538 physical descriptions presented in three surface forms, with a four-class elementary particle task providing six binary pairs for cross-concept checks. At each of the 36 layers three directions are measured at the answer position. The reading direction is the Fisher discriminant, the optimal linear readout. The writing direction is the difference of class means, the contrast the model writes into the stream. The use direction is the class-averaged gradient of the logit difference, the direction to which the decision is locally most sensitive, with the unembedding contrast as a fixed reference.

The main result is a double dissociation between reading and use. The reading direction decodes the decay type almost perfectly, with a held-out AUC of 0.98 to 0.99, yet intervening along it leaves the answer unchanged, while the projection onto the use direction decodes barely above chance in the layers where intervening along it reverses the class margin on most examples. One direction carries the class information without causal effect, and the other carries the causal effect without linearly readable class information. The inertness of reading is not universal but geometric. On the particle pairs, where the reading direction has a larger component along the use direction, the same intervention flips the answer, so causal relevance tracks alignment with use rather than decodability. An exact attenuation identity explains both halves, the within-class dispersion along the use direction masks it from a linear reader and simultaneously turns the optimal probe away from it.

The mechanism behind the writing direction was traced through the network. Its rotation across depth is carried mainly by the MLP blocks, invariantly across three surface forms and six particle pairs, and its norm grows toward the output while the norm of the use direction shrinks, their ratio reaching two to three orders of magnitude. The transcoder features capture the writing direction well above a random null, by factors of 6 to 15, capture the use direction partially, and leave the optimal reading direction near the null. Clustering the causally important features

shows that they detect physical properties of the decay, the charge, the lepton number, the nuclear composition, and the transition type, activating comparably on both classes and entering in order of depth, so the model writes a physical description of the process and assembles the class label only in a narrow decision zone in the late layers. Under steering the writing and use directions dissociate qualitatively, a push along the writing direction moves the continuation coherently toward the paired class, while a push along the use direction forces the class token and degenerates the continuation.

For interpretability practice the consequence is twofold. The decodability of a concept, whether by a probe or in a single dictionary feature, must be paired with a causal test before it is identified with the mechanism the model uses, and the content a model writes need not be the category it predicts but the more elementary properties from which that category is computed. The main limitations are that the results concern one model and two concept families, that the use direction and the feature attributions are first-order objects, and that the feature span analysis establishes expressibility by the active dictionary rather than a realised write.

All experiments are implemented as a documented and tested pipeline of numbered Python scripts with per-experiment outputs, developed and run on the CSD3 high performance computing cluster, and the repository accompanies the report together with the declaration of the use of autogeneration tools.